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New Classes of Perfect Maps II

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The existence and construction of perfect maps, also known as de Bruijn arrays or de Bruijn tori, is further considered. A c-ary $(r, s; u, v)$ perfect map is a two-dimensional periodic array with periods r and s and symbols from an alphabet of size c with the property that every possible $u \times v$ array of symbols occurs exactly once in a period of the array. Necessary conditions on the parameters r, s, u, v for the existence of perfect maps are known to be sufficient when c is a power of a prime. This result is combined with some generalisations of the techniques of Mitchell to construct c-ary perfect maps for a large class of parameter sets. Our results are the strongest yet obtained on the existence question for c-ary perfect maps. © 1996 Academic Press, Inc.

1. INTRODUCTION

We recall the definitions and notations introduced in [6]. Let $c \geqslant 2$ be a positive integer and define $\mathscr{A}$ to be an alphabet of size c, that is, any set with c elements. Unless otherwise stated, we will assume that $\mathscr{A} = \{0, 1, ..., c-1\}$.

Let $A = (a_{i,j})$, $i, j \in \mathbf{Z}$, be an infinite two-dimensional array with period (r, s) over alphabet $\mathscr{A}$ of size c. That is,

$$a_{i,j} = a_{i+r,j} = a_{i,j+s} \qquad \text{for all} \quad i, j \in \mathbf{Z}, \text{ where } a_{i,j} \in \mathscr{A}.$$

The (u, v) subarray of A at position (x, y) is defined to be the $u \times v$ array whose entries are

$$a_{x+i,\, y+j}, \qquad 0 \leqslant i < u; \quad 0 \leqslant j < v.$$

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DEFINITION. We say that A is a c-ary $(r, s; u, v)$ *perfect map* (PM) if every possible c-ary $u \times v$ array occurs as a (u, v) subarray of A in a unique position (x, y) with $0 \leqslant x < r$ and $0 \leqslant y < s$.

The reader is referred to [6] for a summary of what is known about perfect maps. We only restate necessary conditions on the parameters.

RESULT 1.1 [6, Result 1.1]. *Suppose there exists a c-ary $(r, s; u, v)$ PM. Then*

(i) $rs = c^{uv}$,

(ii) $r > u$ *or* $r = u = 1$,

(iii) $s > v$ *or* $s = v = 1$.

The cases where $r = u = 1$ or $s = v = 1$ correspond merely to de Bruijn sequences and need not concern us further here. The case where $c = p^{\alpha}$ is a power of a prime was considered in [6], the main result of that paper being the following.

RESULT 1.2 [6, Theorem 1.4]. *The conditions of Result* 1.1 *are sufficient for the existence of a c-ary $(r, s; u, v)$ PM when c is a power of a prime.*

Suppose now that c has prime factorisation

$$c = \prod_{i=1}^{n} p_i^{\alpha_i}$$

and that $(r, s; u, v)$ is a parameter set satisfying the conditions of Result 1.1. By Result 1.1(i), any prime dividing r or s also divides c. We therefore write

$$r = \prod_{i=1}^{n} p_i^{k_i}, \qquad s = \prod_{i=1}^{n} p_i^{\alpha_i uv - k_i}$$

for some $0 \leqslant k_i \leqslant \alpha_i uv$ with $r > u$, $s > v$.

We conjectured in [6] that the conditions of Result 1.1 are in fact sufficient for the existence of perfect maps. Our main (constructive) result in this paper is the following.

THEOREM 1.3. *Suppose c, r, and s have prime factorisations as above and that for some i we have*

$$p_i^{k_i} > u, \qquad p_i^{\alpha_i uv - k_i} > v.$$

Then there exists a c-ary $(r, s; u, v)$ perfect map.

Clearly the parameter sets satisfying Result 1.1 for which our conjecture remains unsettled are those where (in the notation of Theorem 1.3) either $p_i^{k_i} \leqslant u$ or $p_i^{\alpha_i uv - k} \leqslant v$ for each i. We remark that for fixed u and v, all but a logarithmic fraction of the parameter sets allowed by Result 1.1 are covered under Theorem 1.3. The cases where either $p_i^{k_i} \leqslant u$ for every i or $p_i^{\alpha_i uv - k} \leqslant v$ for every i could be resolved if the existence question for c-ary perfect factors were positively settled (see [2, 3, 5] for recent progress on this problem). However, there would still remain the "mixed cases," where $p_i^{k_i} \leqslant u$ for some indices and $p_i^{\alpha_i uv - k} \leqslant v$ for other indices. It seems that some new techniques are required to construct PMs in these cases.

As an immediate and trivial corollary of Theorem 1.3 we can settle the existence question for square PMs, that is, PMs having $r = s$ and $u = v$. A summary of previous results on square PMs was given in [6, Result 1.2].

COROLLARY 1.4. *There exists a c-ary $(r, r; u, u)$ PM if and only if either u is even or c is a perfect square.*

Proof. From Result 1.1(i), a square PM must have $r = c^{u^2/2}$ an integer, and so the necessity of the conditions is clear. Let c have prime factorisation as above. Sufficiency then follows from an application of Theorem 1.3; if u is even, then $u \geqslant 2$ and for each prime p_i, $\alpha_i \geqslant 1$, and so $p_i^{\alpha_i u^2/2} > 2$. If c is a perfect square, then for all p_i dividing c, we have $\alpha_i \geqslant 2$ and again $p_i^{\alpha_i u^2/2} > 2$. ∎

Of course, Theorem 1.3 encompasses much more than the case of square PMs. We illustrate the power of our main result with an example.

EXAMPLE 1.5. Consider the case $c = 6$, $u = v = 3$. From Result 1.1, a 6-ary $(r, s; 3, 3)$ PM must have $r = 2^{k_1} 3^{k_2}$, where $0 \leqslant k_1, k_2 \leqslant 9$, and

$$(k_1, k_2) \notin \{(0, 0), (0, 1), (1, 0), (8, 9), (9, 8), (9, 9)\}.$$

Thus there are 94 parameter sets allowed by Result 1.1. It is possible to construct PMs for 24 of these sets by using the previous best methods in [5]. On the other hand, using Theorem 1.3, we obtain 6-ary $(r, s; 3, 3)$ of PMs for 84 of the 94 allowed pairs r, s. The 10 unresolved cases are those where

$$(k_1, k_2) \in \{(0, 8), (0, 9), (1, 1), (1, 8), (1, 9), (8, 0), (8, 1), (8, 8), (9, 0), (9, 1)\}.$$

2. Perfect Maps from Perfect Multifactors

We begin by introducing another class of combinatorial objects.

Definition. Let m, n, c, and v be positive integers with $c \geqslant 2$ and $m \mid c^v$. An (m, n, c, v)-perfect multifactor, or simply an (m, n, c, v)-PMF, is a set of c^v/m c-ary sequences of period mn with the property that for every c-ary v-tuple $(t_1, t_2, ..., t_v)$ and for every i with $0 \leqslant i < n$, $(t_1, t_2, ..., t_v)$ occurs at a position $p \equiv i \pmod{n}$ in one of the cycles.

Without loss of generality we can assume that $0 \leqslant p < mn$ in the above definition. A simple counting argument shows that every v-tuple will in fact occur exactly n times in the set of cycles, namely once in each of the possible position congruency classes (mod n). PMFs were first considered by Mitchell in [2], where they were used to construct perfect factors. Necessary conditions that the parameters must satisfy are easily derived. In addition we have the following important constructive result.

Result 2.1 [2, Corollary 4.17]. *Suppose m, n, c, and v are positive integers with $c \geqslant 2$ and $m \mid c^v$. If $n \geqslant v$ then there exists an (m, n, c, v)-perfect multifactor.*

We will consider the analogous two-dimensional objects.

Definition. Let m_1, m_2, n_1, n_2, c, and v be positive integers with $c \geqslant 2$ and $m_1 m_2 \mid c^{uv}$. A two-dimensional $(m_1, m_2, n_1, n_2, c, u, v)$-perfect multifactor is a set of $c^{uv}/m_1 m_2$ two-dimensional c-ary arrays of period $(m_1 n_1, m_2 n_2)$ with the property that for every c-ary $u \times v$ array $T = (t_{i,j})$ and for every i, j with $0 \leqslant i < n_1$, $0 \leqslant j \leqslant n_2$, T occurs at a position (p_1, p_2) with

$$p_1 \equiv i \pmod{n_1}$$

$$p_2 \equiv j \pmod{n_2}$$

in one of the arrays.

Again, a simple counting argument shows that every $u \times v$ array actually occurs exactly $n_1 n_2$ times in the set of arrays—once in each possible position modulo (n_1, n_2). The importance of two-dimensional PMFs to us lies in the next result, a partial generalisation of [2, Construction 5.1].

Lemma 2.2. *Suppose there exists a c-ary $(n_1, n_2; u, v)$ PM and a $(m_1, m_2, n_1, n_2, d, u, v)$-PMF with $m_1 m_2 = d^{uv}$. Then there exists a (cd)-ary $(m_1 n_1, m_2 n_2; u, v)$ PM.*

Proof. Let $A=(a_{i,j})$ be a c-ary $(n_1, n_2; u, v)$ PM. Since $m_1 m_2 = d^{uv}$ the PMF in the theorem consists of just one array, which we denote by $B=(b_{i,j})$. We define a new (cd)-array $C=(c_{i,j})$ by writing

$$c_{i,j}=a_{i,j}+cb_{i,j} \qquad \text{for all} \quad i,j\in\mathbf{Z} \tag{2.1}$$

and claim that C is a (cd)-ary $(m_1 n_1, m_2 n_2; u, v)$ PM.

First, since A has period (n_1, n_2) and B has period $(m_1 n_1, m_2 n_2)$, it follows from (2.1) that C has period $(m_1 n_1, m_2 m_2)$. Now let $T=(t_{i,j})$ be an arbitrary (cd)-ary $u\times v$ array. In view of the fact that $m_1 n_1 \cdot m_2 n_2 = (cd)^{uv}$, we need only show that T occurs as a subarray of C. Let $U=(u_{i,j})$ be the c-ary $u\times v$ array with

$$u_{i,j}\equiv t_{i,j} \pmod{c}$$

and $V=(v_{i,j})$ the d-ary $u\times v$ array with

$$v_{i,j}=(t_{i,j}-u_{i,j})/c.$$

Thus we have $T=U+cV$.

Since A is a c-ary $(n_1, n_2; u, v)$ PM, there exist i, j with $0\leqslant i<n_1$ and $0\leqslant j<n_2$ such that U occurs at position (i, j) in A. From the fact that B is an $(m_1, m_2, n_1, n_2, d, u, v)$-PMF, there exist p_1, p_2 with

$$p_1\equiv i \pmod{n_1}$$
$$p_2\equiv j \pmod{n_2}$$

such that V occurs at position (p_1, p_2) in B. From the periodicity of A we see that U also occurs at position (p_1, p_2) in A. Finally, from the construction of C, in particular (2.1), we can conclude that T occurs at position (p_1, p_2) in C. ∎

This result is capable of significant further generalisation. Suppose we define a two-dimensional (n_1, n_2, c, u, v) perfect factor to be a set of arrays of period (n_1, n_2) having the property that every c-ary $u\times v$ array occurs exactly once as a subarray in one of the arrays. Then an almost identical argument to that used above shows that an (n_1, n_2, c, u, v) perfect factor and an $(m_1, m_2, n_1, n_2, d, u, v)$-PMF can be combined to produce an $(m_1 n_1, m_2 n_2, cd, u, v)$ perfect factor. Two-dimensional perfect factors can themselves be used to produce three-dimensional PMs in a generalisation of Etzion's construction (see [1]). We de not pursue these possibilities further here.

3. Constructions for Two-Dimensional PMFs

A general theory of two-dimensional PMFs is interesting for its own sake and, as indicated above, has importance in the construction of higher

dimensional perfect maps. However, in this paper we will give only two constructions for PMFs that are of direct use in the proof of Theorem 1.3.

First, we give a particularly easy method for constructing a PMF from a PM.

LEMMA 3.1. *Suppose there exists a c-ary* $(m_1, m_2; u, v)$ *PM. Then for any positive integers* n_1 *co-prime to* m_1 *and* n_2 *co-prime to* m_2, *there exists an* $(m_1, m_2, n_1, n_2, c, u, v)$*-PMF.*

Proof. Let $A=(a_{i,j})$ be a c-ary $(m_1, m_2; u, v)$ PM. A has period (m_1, m_2) and therefore also period $(m_1 n_1, m_2 n_2)$. We claim that A is in fact an $(m_1, m_2, n_1, n_2, c, u, v)$-PMF. Let T be an arbitrary c-ary $u \times v$ array and let $0 \leqslant i < n_1$ and $0 \leqslant j < n_2$. We claim that T occurs at a position (p_1, p_2) in A with

$$p_1 \equiv i \pmod{n_1},$$

$$p_2 \equiv j \pmod{n_2}.$$

To verify this claim, note that from the properties of A, there exist q_1, q_2 with $0 \leqslant q_1 < m_1$ and $0 \leqslant q_2 < m_2$ such that T occurs at position (q_1, q_2) in A. Then for every choice of integers x_1, x_2, T also occurs at position $(q_1 + x_1 m_1, q_2 + x_2 m_2)$ in A. However, m_1 is co-prime to n_1 and m_2 is co-prime to n_2, so there exists a choice of (x_1, x_2) such that

$$p_1 = q_1 + x_1 m_1 \equiv i \pmod{n_1}$$

$$p_2 = q_2 + x_2 m_2 \equiv j \pmod{n_2}$$

and T occurs at position (p_1, p_2) in A. ∎

Second, we show how to construct a class of two-dimensional PMFs using the one-dimensional PMFs of Result 2.1.

THEOREM 3.2. *Suppose* $u, v \geqslant 2$ *and* m_1, n_1, *and* n_2 *satisfy* $m_1 \mid d^u$, $n_1 \geqslant u$, *and* $n_2 \geqslant v$. *Suppose further that in the case, where* $v = 2$, m_1 *is even and* n_2 *is odd, we have* $2m_1 \mid d^u$. *Then there exists an* $(m_1, m_2, n_1, n_2, d, u, v)$*-PMF, where* $m_1 m_2 = d^{uv}$.

Proof. Since $m_1 \mid d^u$ and $n_1 \geqslant u$, there exists a one-dimensional (m_1, n_1, d, u)-PMF (see Result 2.1) consisting of d^u/m_1 sequences, which we denote by

$$(s)^0, (s)^1, \ldots, (s)^{(d^u/m_1)-1}.$$

Notice that because $m_1 m_2 = d^{uv}$, our final two-dimensional PMF should consist of just one array. Our aim is to arrange shift of the sequences $(s)^i$

to form the columns of this array in a construction similar to that for PMs given in [4].

The ordering and shifting of the $(s)^i$ is specified in terms of two further PMFs.

Since $n_2 \geqslant v$ and (trivially) $(d^u/m_1)^v \mid (d^u/m_1)^v$, there exists by Result 2.1 an $((d^u/m_1)^v, n_2, d^u/m_1, v)$-PMF, consisting of just one sequence of period $(d^u/m_1)^v n_2$ which we denote by (r). Similarly, there exists an $(m_1^{v-1}, (d^u/m_1)^v n_2, m_1, v-1)$-PMF, consisting of one sequence of period

$$m_1^{v-1}\left(\frac{d^u}{m_1}\right)^v n_2 = \frac{d^{uv}}{m_1} n_2 = m_2 n_2,$$

which we denote by (a). We define a new sequence (w) by writing

$$w_0 = 0, \qquad w_i \equiv \sum_{j=0}^{i-1} a_j \pmod{m_1}, \qquad i \geqslant 1,$$

and define a d-ary array A by taking column i of A to be $E^{n_1 w_i}(s)^{r_i}$. Here, E denotes the left shift operator defined as follows: if $(s) = \ldots, s_{-1}, s_0, s_1, \ldots$ is an arbitrary sequence, then $E(s)$ is the sequence whose ith term is s_{i+1}.

Alternatively, we can think of column i of A as being sequence $(s)^{r_i}$ shifted by a_{i-1} relative to column $i-1$ of A.

We claim that A is an $(m_1, m_2, n_1, n_2, d, u, v)$-PMF. We begin by showing that A has the required period $(m_1 n_1, m_2 n_2)$. We will do this in several steps. First, we show that (w) has period $m_2 n_2$. To this end, we claim that

$$\sum_{j=0}^{m_2 n_2 - 1} a_j \equiv 0 \pmod{m_1}. \tag{3.1}$$

Here the sum includes every term from a generating cycle of (a) exactly once. Since every m_1-ary $(v-1)$-tuple $(t_1, \ldots, t_{v-1})$ occurs exactly $(d^u/m_1)^v n_2$ times in the sequence (a), each j with $0 \leqslant j < m_1$ occurs exactly $m_1^{v-2}(d^u/m_1)^v n_2 = (d^{uv}/m_1^2) n_2$ times in the sequence (a). Thus,

$$\sum_{j=0}^{m_2 n_2 - 1} a_j = \frac{d^{uv}}{m_1^2} n_2 \sum_{j=0}^{m_1 - 1} j = m_1 n_2 (m_1 - 1) \frac{d^{uv}}{2m_1^2}$$

and (3.1) holds if and only if $n_2(m_1 - 1)(d^{uv}/2m_1^2)$ is an integer. Recall that $m_1 \mid d^u$ and $v \geqslant 2$, so that $m_1^2 \mid d^{uv}$. If either n_2 is even or m_1 is odd then also $2 \mid n_2(m_1 - 1)$ and (3.1) holds. Otherwise we have both n_2 odd and m_1 even. We consider two cases. When $v = 2$, the divisibility condition in the statement of the theorem comes into play and (3.1) holds. Otherwise $v \geqslant 3$. Since m_1 is even and divides d^u, we have $2 \mid d^u$. Then $2m_1^2 \mid d^{uv}$ and again (3.1) holds.

From (3.1) we see that, for every $i \geqslant 0$,

$$w_{i+m_2n_2} \equiv w_i + \sum_{j=i}^{m_2n_2+i-1} a_j \equiv w_i + \sum_{j=0}^{m_2n_2-1} a_j \equiv w_i \pmod{m_1}.$$

Thus (w) has period m_2n_2.

Now consider column $i+m_2n_2$ of A; because (r) has a period dividing m_2n_2, this column is simply the shift by $n_1 w_{i+m_2n_2}$ of the sequence $(s)^{r_i}$ of period m_1n_1. But $n_1 w_{i+m_2n_2} = n_1 w_i$. Therefore column $i+m_2n_2$ of A is identical to column i of A. Thus the rows of A have period m_2n_2. Since the columns of A are just shifts of the sequences $(s)^i$ of period m_1n_1, we conclude that A indeed has period (m_1n_1, m_2n_2).

Now let T be an arbitrary d-ary $u \times v$ array and let i, j satisfy $0 \leqslant i < n_1$ and $0 \leqslant j < n_2$. In view of the periodicity of A, we need only show that T occurs in A at a position (p_1, p_2) with

$$p_1 \equiv i \pmod{n_1},$$

$$p_2 \equiv j \pmod{n_2}.$$

For each $1 \leqslant k \leqslant v$, let $\mathbf{c}_k$ denote the vector $(t_{1,k}, t_{2,k}, \ldots, t_{u,k})$ obtained from column k of T. Since the sequences $(s)^0, (s)^1, \ldots, (s)^{(d^u/m_1)-1}$ form an (m_1, n_1, d, u)-PMF, for each $1 \leqslant k \leqslant v$ there exist e_k, f_k such that $\mathbf{c}_k$ is the subsequence of $(s)^{e_k}$ at position f_k and

$$f_k \equiv i \pmod{n_1}.$$

Moreover, since (r) is an $((d^u/m_1)^v, n_2, d^u/m_1, v)$-PMF, there exists a $0 \leqslant x < (d^u/m_1)^v n_2$ such that $(e_1, e_2, \ldots, e_v)$ is the subsequence of (r) at position x and

$$x \equiv j \pmod{n_2}. \tag{3.2}$$

Notice that $n_1 \mid (f_{k+1} - f_k)$ for each $1 \leqslant k \leqslant v-1$. We write

$$g_k = (f_{k+1} - f_k)/n_1.$$

Since (a) is an $(m_1^{v-1}, (d^u/m_1)^v n_2, m_1, v-1)$-PMF, there exists $0 \leqslant p_2 < m_2n_2$ such that $(g_1, g_2, \ldots, g_{v-1})$ is the subsequence of (a) at position p_2 and

$$p_2 \equiv x \pmod{\left(\frac{d^u}{m_1}\right)^v n_2}. \tag{3.3}$$

From this last congruence and the periodicity of (r) we see that the subsequence of (r) at position p_2 is again $(e_1, e_2, ..., e_v)$. In the array A, therefore, columns p_2, p_2+1, ..., p_2+v-1 are just the sequences

$$E^{n_1 w_{p_2}}(s)^{e_1},\ E^{n_1 w_{p_2+1}}(s)^{e_2},\ ...,\ E^{n_1 w_{p_2+v-1}}(s)^{e_v}.$$

But for $k \geqslant 1$,

$$\begin{aligned} w_{p_2+k} &= w_{p_2+k-1} + n_1 a_{p_2+k-1} \\ &= w_{p_2+k-1} + n_1 g_k \\ &= w_{p_2+k-1} + f_{k+1} - f_k \end{aligned}$$

so these columns of A are just the sequences $(s)^{e_1}$, $(s)^{e_2}$, ..., $(s)^{e_v}$ in relative shifts $f_2 - f_1, f_3 - f_2, ..., f_v - f_{v-1}$. But vectors $\mathbf{c}_1, \mathbf{c}_2, ..., \mathbf{c}_v$ occur in sequences $(s)^{e_1}$, $(s)^{e_2}$, ..., $(s)^{e_v}$ in precisely these relative shifts. So A contains T as a subarray at position (p_1, p_2), where p_1 is the position of $\mathbf{c}_1$ in column p_2 of A, that is, $E^{n_1 w_{p_2}}(s)^{e_1}$. Therefore,

$$p_1 = -n_1 \sum_{j=0}^{p_2-1} a_j + f_1 \equiv i \pmod{n_1}.$$

Since $n_2 \mid (d^u/m_1)^v n_2$, Eqs. (3.2) and (3.3) give

$$p_2 \equiv x \equiv j \pmod{n_2}.$$

Hence A contains T as a subarray at a position (p_1, p_2) with

$$\begin{aligned} p_1 &\equiv i \pmod{n_1}, \\ p_2 &\equiv j \pmod{n_2}. \end{aligned}$$ ∎

We note that $(m_1, m_2, n_1, n_2, d, u, 1)$-PMFs with $u \geqslant 1$, $n_1 \geqslant u$, $m_1 \mid d^u$, and $m_1 m_2 = d^u$ can be easily constructed by appropriately ordering the sequences of a one-dimensional (m_1, n_1, d, u)-PMF to form the columns of an array.

4. Proof of Theorem 1.3

We have now assembled enough tools to prove our main result. We consider an alphabet of size $c = \prod_{i=1}^{n} p_i^{\alpha_i}$ and write

$$r = \prod_{i=1}^{n} p_i^{k_i}, \qquad s = \prod_{i=1}^{n} p_i^{\alpha_i uv - k_i},$$

where we have

$$p_i^{k_i} > u, \qquad p_i^{\alpha_i uv - k_i} > v$$

for some i. For convenience, we write $l_i = \alpha_i uv - k_i$, $1 \leqslant i \leqslant n$. We aim to construct a c-ary $(r, s; u, v)$ PM.

Let G_1 be the set of indices i for which $p_i^{k_i} \leqslant u$, G_2 the set of indices i for which $p_i^{l_i} \leqslant v$ and G_3 the remaining indices. Notice that G_1, G_2, and G_3 are disjoint sets with G_3 nonempty.

We begin by dealing with the case where $G_1 = G_2 = \varnothing$. Here we have

$$p_i^{k_i} > u, \qquad p_i^{l_i} > v \qquad \text{for every} \quad 1 \leqslant i \leqslant n$$

and so by Result 1.2, there exists a $p_i^{\alpha_i}$-ary $(p^{k_i}, p^{l_i}; u, v)$ PM, A_i, for each i. Using A_2 in Lemma 3.1, we can construct a $(p_2^{k_2}, p_2^{l_2}, p_1^{k_1}, p_1^{l_1}, p_2^{\alpha_2}, u, v)$-PMF consisting of one array B_2. We combine A_1 and B_2 to produce a $(p_1^{\alpha_1} p_2^{\alpha_2})$-ary $(p_1^{k_1} p_2^{k_2}, p_1^{l_1} p_2^{l_2}; u, v)$ PM using Lemma 2.2. Repeatedly using Lemma 3.1 and Lemma 2.2 in this way, it is easy to see how a $(\prod_{i=1}^n p_i^{\alpha_i})$-ary $(\prod_{i=1}^n p_i^{k_i}, \prod_{i=1}^n p_i^{l_i}; u, v)$ PM can be constructed.

Now suppose that $G_2 = \varnothing$ but G_1 is nonempty. We write

$$c = \prod_{i \in G_3} p_i^{\alpha_i}, \qquad d = \prod_{i \in G_1} p_i^{\alpha_i},$$

$$n_1 = \prod_{i \in G_3} p_i^{k_i}, \qquad n_2 = \prod_{i \in G_3} p_i^{l_i},$$

$$m_1 = \prod_{i \in G_1} p_i^{k_i}, \qquad m_2 = \frac{d^{uv}}{m_1}.$$

As above we can construct a c-ary $(n_1, n_2; u, v)$ PM A. Further we claim that there exists an $(m_1, m_2, n_1, n_2, d, u, v)$-PMF. We aim to apply Theorem 3.2. Since G_3 is nonempty and $p_i^{k_i} > u$ for each $i \in G_3$, we immediately have $n_1 \geqslant u$ and $n_2 \geqslant v$. Moreover, since for each $i \in G_1$ we have $p_i^{k_i} \leqslant u < p_i^{\alpha_i u}$, it follows that $k_i < \alpha_i u$ for each $i \in G_1$. Thus $m_1 \mid d^u$ and, when m_1 is even, $2m_1 \mid d^u$. From Theorem 3.2 (or the note following that theorem when $v = 1$), there exists an $(m_1, m_2, n_1, n_2, d, u, v)$-PMF consisting of one array B. Using the PM A and PMF B together in Lemma 2.2, we can finally construct a (cd)-ary $(m_1 n_1, m_2 n_2; u, v)$ PM which has the desired parameters.

The final case is when G_2 is also nonempty and can be dealt with as follows. This time we write

$$c = \prod_{i \in G_1 \cup G_3} p_i^{\alpha_i}, \qquad d = \prod_{i \in G_2} p_i^{\alpha_i},$$

$$n_1 = \prod_{i \in G_1 \cup G_3} p_i^{l_i}, \qquad n_2 = \prod_{i \in G_1 \cup G_3} p_i^{k_i},$$

$$m_1 = \prod_{i \in G_2} p_i^{l_i}, \qquad m_2 = \frac{d^{uv}}{m_1}.$$

Using the procedure described above we can construct a c-ary $(n_2, n_1; u, v)$ PM and so, by transposing, a c-ary $(n_1, n_2; v, u)$ PM, A. Similar reasoning to that above also shows that there exists an $(m_1, m_2, n_1, n_2, d, v, u)$-PMF consisting of one array B. Combining A and B in an application of Lemma 2.2, we can finally construct a (cd)-ary $(m_1 n_1, m_2 n_2; v, u)$ PM whose transpose has the desired parameters. The proof of Theorem 1.3 is now complete. ∎

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REFERENCES

1. T. ETZION, Constructions for Perfect Maps and pseudo-random arrays, *IEEE Trans. Inform. Theory* **IT-34** (1988), 1308–1316.
2. C. J. MITCHELL, Constructing c-ary Perfect Factors, *Designs Codes & Cryptography* **4** (1994), 341–368.
3. C. J. MITCHELL, New c-ary Perfect Factors in the de Bruijn graph, *in* "Codes and Cyphers—Cryptography and Coding IV" (P. G. Farrell, Ed.), IMA Press, Southend-on-Sea, Essex, UK, 1995.
4. C. J. MITCHELL AND K. G. PATERSON, Decoding Perfect Maps, *Designs Codes & Cryptography* **4** (1994), 11–30.
5. K. G. PATERSON, Perfect Factors in the de Bruijn graph, *Designs Codes & Cryptography* **5** (1995). 115–138.
6. K. G. PATERSON, New classes of Perfect Maps I, *J. Combin. Theory Ser. A* **73** (1996), 302–334.